

The tadpole reabsorption lemma: normal-ordered cubic vertices in the matched bookkeeping eliminate the tadpole channel identically

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

Unit U4 identified the tadpole channel ($9M^2\hat{G}$ of $\langle V_3^2 \rangle_c$) as load-bearing for any SC-SCOPE lift. This note proves the channel is absent identically in the matched bookkeeping, discharging the v1.0 residuals R-U6-1 (this writeup) and R-U6-2 (the machine cross-check). **Scope:** the matched bookkeeping of the B5 chain (per-pattern self-consistent Hartree dressing + per-pattern condensate optimization, Math426–Math437). **Not claimed:** any change to the sunset channel, any SC-SCOPE margin, any tier or gate.

2. The normal-ordering identity

Let δ be the Gaussian fluctuation with variance $M = \langle \delta^2 \rangle$ and let $: \cdot :$ denote Wick normal-ordering with respect to it. The classical identity

$$\delta^n = \sum_{k \geq 0} \binom{n}{2k} (2k-1)!! M^k : \delta^{n-2k} :$$

(one-line proof: apply $\langle e^{\lambda\delta} \rangle = e^{\lambda^2 M/2}$ to $e^{\lambda\delta} = e^{\lambda^2 M/2} : e^{\lambda\delta} :$ and match powers) gives in particular $\delta^3 = 3M : \delta^3 :$, $\delta^4 = 6M : \delta^2 : + 3M^2 + : \delta^4 :$, $\delta^5 = 10M : \delta^3 : + 15M^2 : \delta : + : \delta^5 :$, $\delta^6 = 15M : \delta^4 : + 45M^2 : \delta^2 : + 15M^3 + : \delta^6 :$. All coefficients are machine-verified by exact pairing counts (`ru61_tadpole_alignment.py`, assert 2).

3. Exact alignment with the production bookkeeping

Write $\varphi = c + \delta$ with condensate $c = t\psi$, $\langle \psi^2 \rangle = 1$, and expand the interaction $\frac{u}{4}\varphi^4 + \frac{v}{6}\varphi^6$ by the identity above, collecting by the normal-ordered degree j :

$j = 0$ (**the layer functional**). The c^4 coefficient is $\frac{u}{4} + \binom{6}{2} \frac{v}{6} M \cdot \langle \psi^4 \rangle$ -normalised = $\frac{u}{4} + \frac{5v}{2} M = \frac{u_{\text{eff}}(M)}{4}$ with $u_{\text{eff}} = u + 10vM$: EXACTLY the $0.25 UN_4 + 2.5 VN_4 M$ line of the production engine (Math437 §1). Machine: assert 3 (equality to 10^{-15} at $M = M_R$).

$j = 1$ (**the source**). The coefficient of $: \delta :$ linear in c is $3uM + 15vM^2$ (from δ^3 and δ^5): EXACTLY the gap/stationarity dressing – at the $A = 0$ gap point the production solver gives $m_R - r = 0.29952571$, reproduced by $3uM_R + 15vM_R^2$ to 5×10^{-7} (assert 4, independent solver). The c^3 source is $u_{\text{eff}} t^3$ and the c^5 source the sextic line: together the $j = 1$ collection IS $\partial\Phi/\partial t$ plus the $\hat{\kappa}$ dressing, which VANISHES at each pattern’s own optimum – the matched bookkeeping’s defining property.

$j = 2$. The $: \delta^2 :$ collection is the per-pattern gap dressing (the production `gap_solve` line $\hat{r} = r + 3uM + 15vM^2 + 6unA^2 + 60vnA^2M + 5vN_4A^4$); resummed into G .

$j = 3$ (**the cubic vertex**). What remains is

$$V_3 = g_3 : \delta^3 :, \quad g_3 = u_{\text{eff}}(M) c + \frac{10v}{3} c^3,$$

normal-ordered BY CONSTRUCTION – not by choice: the non-normal-ordered pieces of the cubic are precisely the $j = 1$ source already absorbed by stationarity.

4. The lemma (proof)

Lemma (tadpole reabsorption). *In the matched bookkeeping the third-cumulant correction of any competitor pattern is $\delta F_3 = -\frac{1}{2}\langle :V_3:^2 \rangle_c = -\frac{1}{2}\sum_t |\hat{g}_3(t)|^2 6\hat{G}^3(t)$: the sunset channel only; the tadpole contraction $9M^2\hat{G}$ is absent identically.*

Proof. By Sec. 3 the residual cubic vertex is $:\delta^3:$. Wick’s theorem for normal-ordered insertions forbids self-contractions; the brute-force enumeration of the 15 perfect matchings of the six legs of $\langle \delta^3(x)\delta^3(y) \rangle$ splits them as 6 cross-only ($=3!G(x-y)^3$, the sunset) + 9 containing a self-pair (the $9M^2G$ tadpole class) – assert 1 – and exactly the 9 self-pair matchings are the ones killed by normal-ordering. Hence $\langle :\delta^3(x)::\delta^3(y): \rangle = 6G(x-y)^3$ and the tadpole channel does not occur. $\square$

5. The two v1.0 objection cases, written out

(β) Off-optimum competitors. The $j = 1$ source vanishes at each pattern’s OWN layer optimum, which is where the chain’s comparison functional evaluates it (the races optimize (t, m) per pattern; Prop-A quantifies over all trials, dominating off-optimum evaluations from below). For a hypothetical off-optimum evaluation the residual linear term generates an in-layer shift whose energy gain is second-order along the shift direction – an amplitude move inside the Prop-A trial quantification, or a profile move inside the class quantification. No tadpole CONTRACTION arises in either case; only the source re-appears, and it is dominated as above.

(γ) Resonant $O(I^2)$ accounting. The $O(c^3)$ part of g_3 (weight $\frac{10v}{3}t^3$) shifts the stationarity source at relative order $(10v/3)t^2/u_{\text{eff}} = 8.0 \times 10^{-3}$ at the endpoint ($= O(I)$, assert 8); the optimum moves by $O(I)$ and the energy by $O(I^2) \times \text{curvature}$ – which is the N_4 resonance accounting ALREADY inside the matched second-cumulant functional (Math437 gallery). The genuinely new remainder is $O(I^4)$: $4.3 \times 10^{-10} < 10^{-6}$ at the endpoint (assert 8).

6. Self-caught correction of the v1.0 mechanism wording

v1.0 wrote: “the would-be tadpole linear source $3Mg_3 = 3Mu_{\text{eff}}F$ coincides term-by-term with the stationarity-equation source.” This is IMPRECISE: with the dressed g_3 , $3Mu_{\text{eff}} = 3uM + 30vM^2$, whereas the exact $j = 1$ source is $3uM + 15vM^2$ – the naive contract-the-dressed-vertex route counts the double self-loop twice (symmetry factor 2), over-counting by exactly $15vM^2 = 0.58181468$ at M_R (assert 6). The CORRECT mechanism is the Hermite collection of Sec. 3, in which each self-contraction is counted once with its proper factor. The lemma’s conclusion is unaffected – the channel is absent either way – but the v1.0 shorthand would have failed a referee’s coefficient check, so it is corrected here rather than silently rewritten.

7. Numerical verification

codes/vacuum/ru61_tadpole_alignment.py (8/8 asserts PASS, exit 0; JSON artefact runs/260612-ru61-tadpole-alignment/result.json): pairing split $15 = 6 + 9$ (brute force); Hermite coefficients exact (integer combinatorics + moment closure); $j = 0$ quartic $= u_{\text{eff}}/4$ to 10^{-15} ; $j = 1$ source $= m_R - r$ against the independent production gap solver; Math437 Lemma-1 boxed identity $u_{\text{eff}}(M_+) = \frac{1}{3}\sqrt{9u^2 - 60vr}$ to 10^{-12} ; the $15vM^2$ over-count quantified; the Math436/Math437

HEX argmin $\hat{r} = 0.3093733$ reproduced (clearly-labelled oracle); the endpoint remainder 4.3×10^{-10} ; $u_{\text{eff}}(M_R) = 2.685$ matching the certified value.

Quantitative sanity check (mandatory): direction – striking a negative-definite channel can only raise competitor free energies (selection-favorable), fixed by combinatorics, not magnitude tuning; limit case – for $\mathcal{R}_H$ ($c = 0$) both channels vanish and the lemma is trivial; the corrected-vs-naive source discrepancy ($15vM^2$) is POSITIVE, i.e. the naive shorthand was conservative in the dangerous direction (it over-stated what stationarity absorbs), so no prior conclusion leaned on the imprecision.

8. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*The Hermite identity is single-mode; the field is a continuum with pattern structure.*” DISMISSED: the identity is applied to the Gaussian fluctuation field mode-wise with M the cell-averaged variance – precisely the production engine’s own Wick bookkeeping (Math424-AddA), whose N_4 lines the $j = 0/j = 1$ collections reproduce EXACTLY (asserts 3–4); the alignment is verified against the engine, not assumed.
- (β) “*v1.0’s mechanism error suggests other coefficients may be wrong too.*” VALID with mitigation: that is why v1.1 machine-verifies EVERY coefficient it uses (asserts 2–6) including the production cross-checks (independent gap solver; Lemma-1 identity; HEX argmin oracle); the one imprecision found is documented in Sec. 6, not patched silently.
- (γ) “*The off-optimum case (Sec. 5 β) leans on Prop-A’s quantification, which is $\mathcal{A}_{\text{adm}}$ -scoped.*” VALID with mitigation: for the C_{full} statement the head package’s own quantification covers the trials; the lemma itself is bookkeeping-level and pattern-agnostic – the only pattern input is WHERE the functional is evaluated, and every chain evaluation point is a per-pattern optimum by construction.

9. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / tadpole-reabsorption-lemma (unit U6; gate R-U6-1; task T-004)
Precise statement:	PROOF (was T3 sketch): in the matched bookkeeping the residual cubic vertex is normal-ordered by construction (Hermite collection: $j=0$ reproduces the $0.25U+2.5VM$ line, $j=1$ the $3uM+15vM^2$ gap dressing, $j=3$ gives $g_3 = u_{\text{eff}} c + (10v/3)c^3$); hence $\langle d^3 : d^3 \rangle = 6G^3$ and the tadpole channel $9M^2G$ is ABSENT IDENTICALLY. U4 tadpole rows struck; sunset is the sole surviving third-cumulant threat. SELF-CAUGHT v1.1 correction: the v1.0 “3M g_3 ” shorthand over-counts the v-line by $15vM^2$; the Hermite route is the exact mechanism.
Scope:	matched bookkeeping of the B5 chain; all anchors. No change to the sunset channel, margins, tiers, or gates.
Evidence grade:	ANALYTIC (written proof, operator-ACCEPTED) + EXECUTED (ru61_tadpole_alignment.py 8/8, exit 0; operator clean-run CONFIRMED). Gates R-U6-1 and R-U6-2 DISCHARGED (operator verdicts 2026-06-12).
Reproduction command:	python codes/vacuum/ru61_tadpole_alignment.py
Falsifier:	a matched-bookkeeping derivation exhibiting a non-cancelling tadpole remainder at $O(I^2)$ or larger beyond the resonant accounting; any

coefficient assert failing (8 registered).

Tier before / after: T3 sketch -> ACCEPTED WRITTEN PROOF
(Tadpole-Reabsorption-Lemma-RU61-RU62-260612);
gates R-U6-1/R-U6-2 DISCHARGED. B5 stays T5
(next tier question = T-031 decision layer).

No-overclaim statement: does NOT lift SC-SCOPE (the sunset endpoint
refinement remains the open item); does not
change any margin; the off-optimum case rests
on the chain's own quantification structure.

Next required action: none for this lemma (gates closed). Mainline:
T-031 exhaustiveness decision layer; the
sunset endpoint remains the binding
third-cumulant channel.